The effects of Immigration Enforcement on Health Care Utilization: A National Analysis

Naomi Buell Richie Rivera

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1 Abstract

Background: The removal of “sensitive location” protections and increased ICE activity function can discourage people from seeking medical care by creating fear, stress, and other barriers to accessing health services, including among immigrant communities not directly targeted by enforcement.

Objective: To assess whether increased ICE enforcement correlates with reduced health care utilization.

Methods: We use national datasets on ICE detention activity and health care utilization to analyze the relationship between enforcement intensity and health care utilization across geographies and over time. We will control for potential confounders such as socioeconomic status, demographic characteristics, and local health care infrastructure.

Results: [TBD]

Conclusions: We found that immigration enforcement [is/is not] associated with reduced health care utilization. [Implications for public health and policy TBD]

2 Introduction

Immigration-related fears appear to be negatively affecting health care access in the United States (Kaiser Family Foundation 2025). Recent survey evidence indicates that nearly half (48%) of likely undocumented immigrant adults and 14% of immigrant adults overall report that they or a family member have avoided seeking medical care due to immigration-related concerns. These findings suggest that immigration enforcement may function as a structural barrier to health care access, even for individuals who are not directly targeted by enforcement actions. In January 2025, federal policy changes reversed prior protections that designated hospitals and other health care facilities as "sensitive locations," limiting enforcement activity in these settings (Department of Homeland Security 2025). The removal of these protections may plausibly increase fears associated with seeking care, particularly in communities with visible or intensified U.S. Immigration and Customs Enforcement (ICE) activity.

Motivated by these developments, this project investigates the relationship between immigration enforcement intensity and health care utilization. Specifically, we seek to answer the question: *Does increased ICE enforcement correlate with reduced health care utilization?* We hypothesize that higher levels of local enforcement activity are associated with greater health care avoidance, as individuals may be less willing to leave their homes or engage with formal institutions perceived as potential sites of immigration enforcement.

2.1 Background

Empirical research provides strong evidence that immigration enforcement activity reduces health care utilization among immigrant populations. At the state level, Hispanic patients were significantly less likely to report having a regular provider or annual checkup following increased ICE activity, a pattern not observed among non-Hispanic adults (Friedman and

Venkataramani 2021). This differential response underscores how enforcement intensity functions as a racialized barrier to care. A more dramatic illustration comes from research on Secure Communities, an immigration enforcement program expanded during the Obama administration from 2008 to 2014, which found that likely authorized Hispanic immigrants experienced a 16.9% decline in the probability of visiting a health care provider compared to non-Hispanic U.S.-born respondents (Herring and Barnow 2025). Complementing these state-level findings, a survey of Asian and Latinx immigrants in California demonstrated that each additional direct experience with immigration enforcement was associated with delayed care for both groups (Young et al. 2023).

Beyond individual-level utilization changes, enforcement has broader structural impacts on health care access. Studies using geographic measures of enforcement intensity, such as ICE detainer requests at the county level, found these requests were associated with lower Medicaid and SNAP enrollment—indicating that enforcement creates barriers to accessing public benefits and formal health care systems (Kravitz et al. 2024). At the regional level, Latino adults living in areas with the most detainer requests had 24% higher odds of fair/poor self-rated health compared to those in areas with the least requests, reflecting enforcement's effects on health outcomes through reduced care access (Eastus et al. 2025). These findings align with a comprehensive literature review synthesizing 32 studies, which documented that hostile immigrant environments correlate with decreased or delayed utilization of basic health care services (Vernice et al. 2020).

The mechanisms linking enforcement to reduced care-seeking are increasingly well-understood through provider and qualitative research. Surveys of clinicians consistently show enforcement's disruptive effects: 48% of primary care and emergency medicine providers reported observing negative effects of ICE enforcement on their immigrant patients' health and health access (Hacker et al. 2012), while another survey found that post-2016 anti-immigrant policies and enforcement actions led half of clinicians to report persistent fear among their immigrant patients—an effect more pronounced in border states (Saadi et al. 2021). Qualitative investigations reveal the emotional and structural pathways through which enforcement affects care-seeking. A 2009 study in Everett, Massachusetts found that enhanced enforcement impacted immigrant health through both emotional means (deportation fears, concerns about ICE-police cooperation) and structural barriers (anxiety about documentation requirements for enrollment and care) (Hacker et al. 2011). An immigration raid in a Midwestern Latino community similarly demonstrated that enforcement actions increase immigration-related stress and lower self-rated health among affected residents (Lopez et al. 2017).

While prior studies have well-documented enforcement effects through state-level analyses, provider surveys, and local case studies, comprehensive research examining the relationship on a national scale at granular geographic and temporal levels remains limited. This project addresses this gap by using national datasets on ICE detention activity and health care utilization to analyze the enforcement-utilization relationship across geographies and over time, allowing us to quantify a relationship that prior research has established in principle but not comprehensively measured at scale.

3 Methods

3.1 Data

We utilize data on immigration enforcement, health care utilization data, and social determinants of health. Detention Facilities Average Daily Population data is available from the Transactional Records Access Clearinghouse (TRAC) and provides information on the average daily population of ICE detention facilities, broken down by city, state, ZIP code, and date from September 2019 through present (Transactional Records Access Clearinghouse 2026). This dataset will allow us to measure the intensity of local immigration enforcement activity over time and across geographies.

Routine checkup within the past year among adults data is available from CDC's PLACES data and provides information on the percentage of adults who reported having a routine checkup within the past year, broken down by county, ZIP Code Tabulation Area (ZCTA), and year from 2020-2025 (Centers for Disease Control and Prevention 2025). This dataset will serve as our primary outcome variable, allowing us to assess health care utilization patterns in relation to local enforcement intensity.

We will use data from the American Community Survey (ACS) 5-Year Estimates to control for social determinants of health in our analysis (U.S. Census Bureau 2018-2023). This data provides information on social determinants of health at the ZCTA level. Social determinants of health analyzed included race, citizenship status, income, education, employment status, and housing conditions.

We used Python to build panel data across datasets, geographies, and years. Our panel data spans from 2018 to 2023 and includes 32,977 unique ZCTAs across the United States. Table 1 provides descriptive statistics for our key variables of interest, including the average daily population in ICE detention facilities (ADP), the percentage of adults reporting a routine checkup within the past year, and key social determinants of health across ZCTAs. The median of our main outcome variable, 76.4%, suggests that for the majority of the

Table 1: Descriptive statistics

Variable	Mean	SD	Min	Median	Max
Percent of adults who had a visit with a doctor or other health care professional in the past year	76	4.7	45	76	96
Percent of population in ICE detention facilities	0.00014	0.0084	0	0	1.8
Miles to nearest detention facility	59	138	0	38	5227
Has ICE facility	0.0048	0.069	0	0	1
Hispanic/Latino	0.097	0.16	0	0.036	1
White	0.81	0.22	0	0.9	1
Black	0.076	0.16	0	0.01	1
Asian	0.023	0.058	0	0.0034	1
Two or more races	0.044	0.058	0	0.028	1
Other race	0.046	0.11	0	0.012	1
Non-citizen	0.031	0.055	0	0.0096	1
Household income	62738	32623	0	58977	250001
Below poverty line	0.13	0.11	0	0.11	1
Unemployed	0.054	0.061	0	0.043	1
High school education or higher	0.89	0.096	0	0.91	1
Without internet access	0.16	0.13	0	0.14	1
Without a vehicle	0.062	0.093	0	0.04	1

United States, annual healthcare engagement is a standard behavior. For most areas of the United States, there are no detention centers (median ADP is 0); however, the maximum ADP reaches 6,280.8, demonstrating that certain areas have a large population in detention facilities. See Appendix A for detailed Python code used in data processing and analysis.

To prepare the data for analysis, we used median imputation to handle missingness, log transformed data to address skewness, and feature engineered our main independent variable of interest, ICE enforcement intensity, by creating a binary variable indicating the presence of an ICE detention facility in a ZCTA code (i.e., whether the average daily population in detention facilities is greater than 0). We also created a continuous variable measuring the distance from each ZCTA code to the nearest ICE detention facility. These two variables allow us to capture both the presence and proximity of enforcement activity, which may have distinct effects on health care utilization.

3.2 Modeling

To analyze the relationship between immigration enforcement intensity and health care utilization, we used a a two-way fixed effects (TWFE) panel regression model. We controlled for the following potential confounding socioeconomic and demographic factors: race, citizenship status, poverty status, high school education, employment status, and housing conditions (rent burden, internet and vehicle access). By leveraging panel data from 2018 to 2023, the TWFE model includes ZCTA code-level fixed effects to absorb all time-invariant geographic and demographic heterogeneity. Additionally, year fixed effects are included to control for macro-level shocks affecting all ZCTA codes simultaneously, most notably the

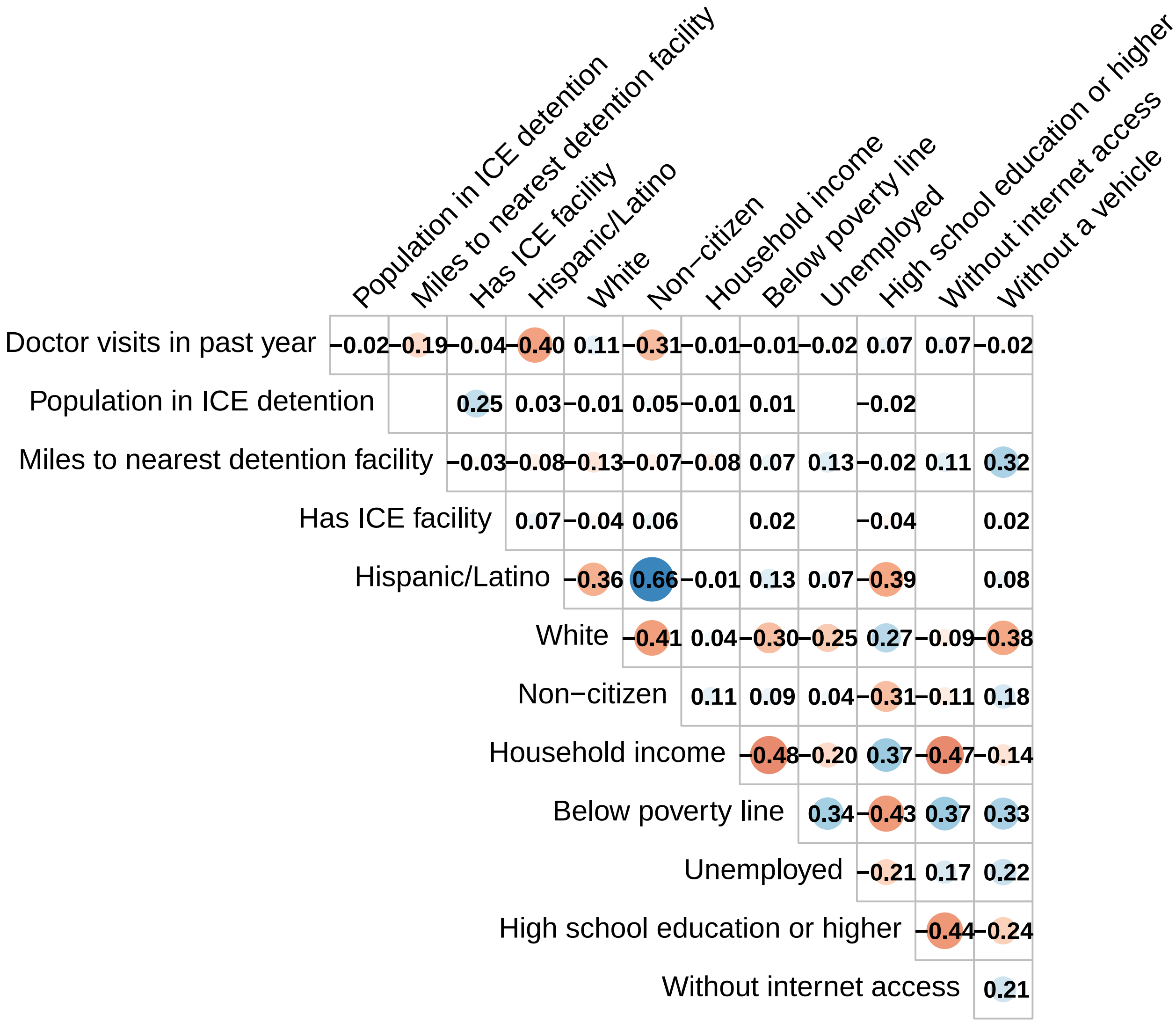

Figure 1: Correlation matrix. Note: Correlations below 0.01 are suppressed.

COVID-19 pandemic and national federal policy shifts. The resulting coefficients represent the within-ZCTA variation: how changes in ICE presence or distance relative to a ZCTA code's own baseline affect the local rate of routine doctor visits. Standard errors are clustered at the ZCTA code level to account for serial autocorrelation.

We built and assessed three different models to capture different dimensions of enforcement intensity:

1. **Model 1: ICE Presence** - This model includes the binary variable indicating the presence of an ICE detention facility in a ZCTA code as the main independent variable, controlling for confounders and fixed effects.

2. **Model 2: Distance to Nearest Facility** - This model includes the continuous variable measuring the distance from each ZCTA code to the nearest ICE detention facility as the main independent variable, controlling for confounders and fixed effects.

3. **Model 3: Regional Chilling Effect** - This model includes a binary variable indicating whether a ZCTA code is within 50 miles of an ICE detention facility as the main independent variable, controlling for confounders and fixed effects.

We assessed model fit using within R-squared.

3.3 Results

Figure 1 provides a visual representation of the correlations between a subset of our analysis variables. The size of the population in ICE detention facilities does not seem to be strongly correlated with the rate of doctor visits in a particular area (r = -0.02). Hispanic/Latino ethnicity and citizenship status are much stronger correlates with doctor visits (r = -0.4 and r = -0.31, respectively, suggesting that Hispanic/Latino folks and non-citizens are less likely to have had a recent doctor's visit). The distance to the nearest ICE detention facility has a slight negative correlation with doctor visits (r = -0.19), which may suggest that proximity to enforcement activity could be associated with reduced health care utilization, although this relationship is not strong.

We visualized a 2023 snapshot geographic distribution of health care utilization and ICE detention centers across the U.S. in Figure 2. This map shows that rates of adults' annual doctor visits vary across the country, with higher rates in the southeast, and lower rates in the west, although it should be noted that there is significant missingness in the west of the country. The spatial distribution of ICE detention centers (in red) illustrates the concentrated nature of federal enforcement infrastructure, particularly along the southern border.

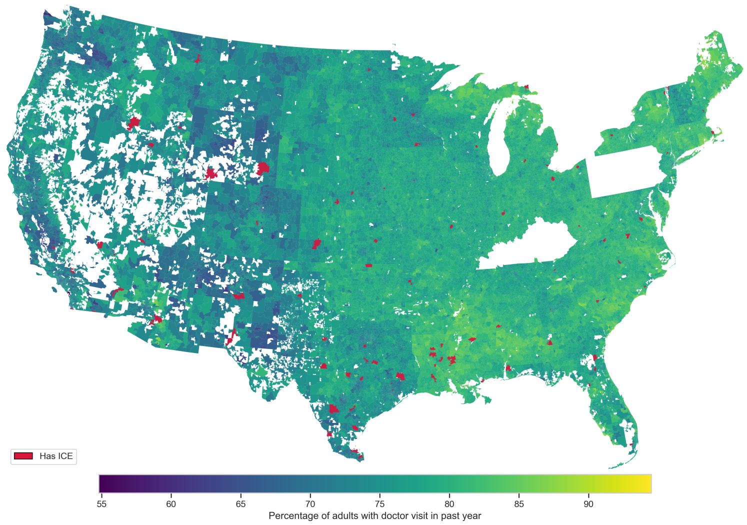

Figure 2: Snapshot of rates of adults who reported a doctor visit within the past year and ICE detention centers, 2023. Note: Some ZCTAs are missing from the data and are left blank.

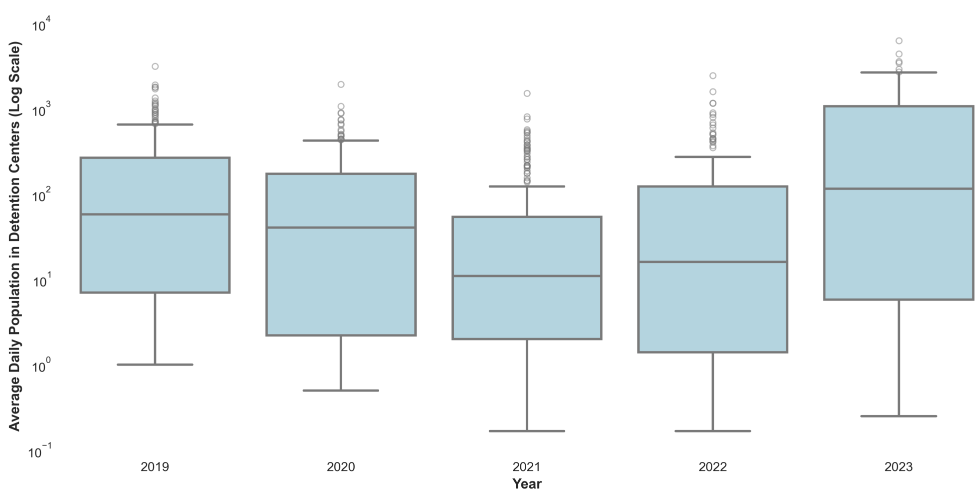

Figure 3: Box plot of average daily population in detention facilities by year

Table 2: Model Summary Results (N = 185,504)

Model	Coefficient (95% CI)	P-value	R-squared (Within)
Model 1: Binary Presence	-0.177 (-0.399, 0.046)	0.1190945	-0.0081636
Model 2: Log Distance	-0.571*** (-0.622, -0.521)	0.0000000	-0.1661899
Model 3: Regional Presence (<=50 Miles)	0.785*** (0.708, 0.862)	0.0000000	-0.1622207

Figure 3 shows that population in detention facilities dropped after 2020, but has been increasing since 2021. The interquartile range also appears to widen in the later years, suggesting that while many ZCTAs continue to have low ADP, a growing number of ZCTAs are experiencing higher levels of enforcement activity.

Table 2 provides a summary of the results from our three regression models.

4 Discussion

Limitations:

5 Conclusion

We conclude that..

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